

Do random slopes matter?

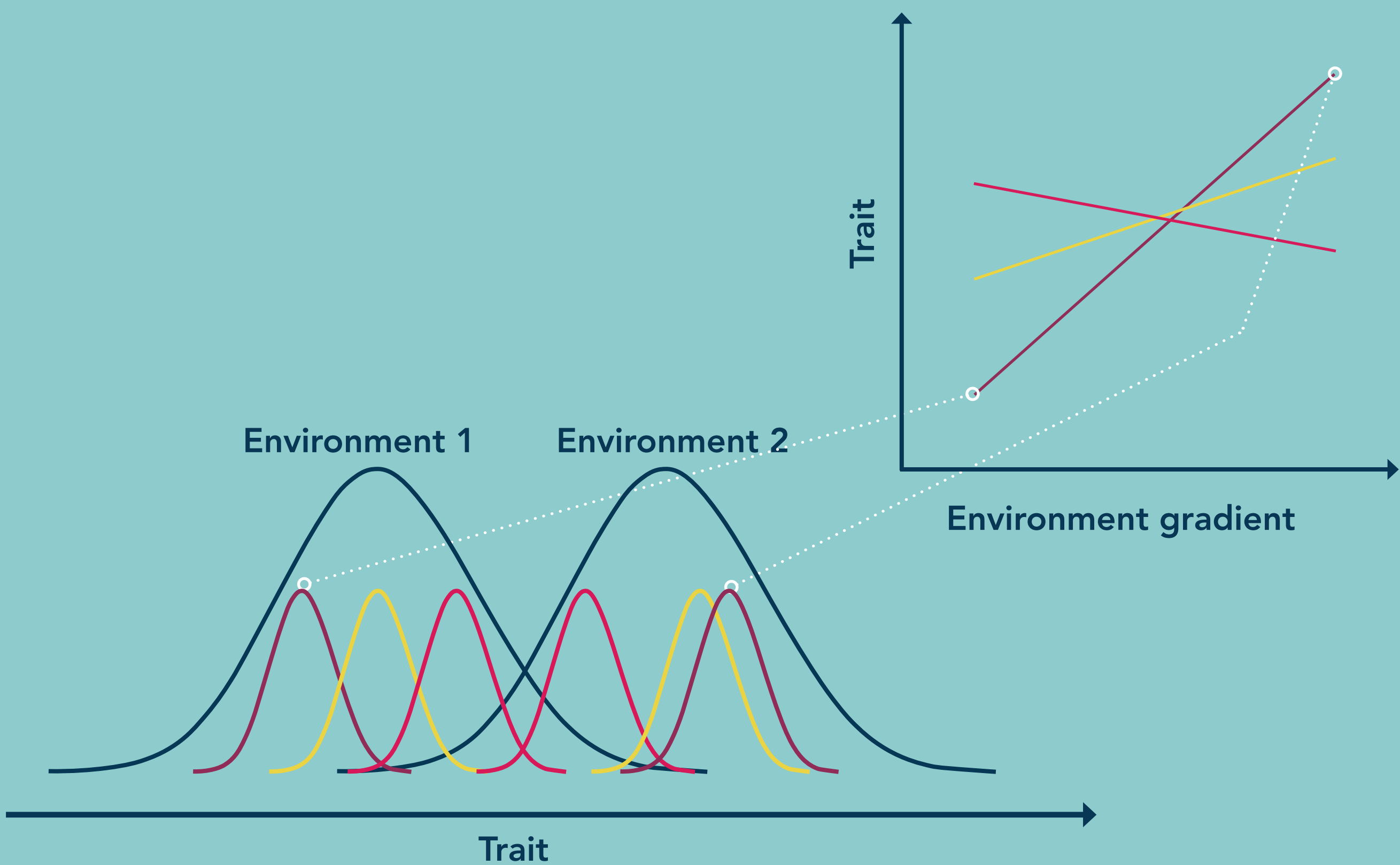
Estimating the importance of among-individual variation in reaction norms across species: a meta-analytic approach

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INTRODUCTION

Reaction norms: Range of possible expression exhibited by an individual organism in response to varying environmental condition.

lme4:lmer(trait ~ 1 + environmental_gradient + (environmental_gradient|individual_ID))



WHAT IS THE MAGNITUDE OF RANDOM SLOPE VARIATION IN NATURAL POPULATION?

$$R_s^2 = \frac{V_s}{V_p} = \frac{V_v V_x + \mu V_v}{V_F + V_I + V_\phi + V_R}$$

V_s : Random slope variance
 V_p : Total variance in response
 μ : Mean of the covariate from the data

V_v : Random slope variance from the model
 V_x : Variance in covariate from the raw data

WHAT IS THE EFFECT OF MODELLING RANDOM SLOPE VARIATION AMONG INDIVIDUALS ON THE POPULATION-LEVEL EFFECT ESTIMATION?

$$\log RR = \frac{\beta_{\text{intercept only}}}{\beta_{\text{intercept+slope}}}$$

SYSTEMATIC REVIEW

IDENTIFICATION

WEB OF SCIENCE = 1966 records
Forward search on four key papers on the topic:
• Schielzeth and Forstmeier 2009 = 508 records
• Dingemanse et al. 2010 = 842 records
• Dingemanse and Dochtermann 2013 = 595 records
• Mitchell et al. 2016 = 21 records

Excluded after de-duplication
200 records

SCREENING

ABSTRACT SCREENING

RECORDS BEFORE ABSTRACT SCREENING = 1766 records
RECORDS AFTER ABSTRACT SCREENING = 1191 records

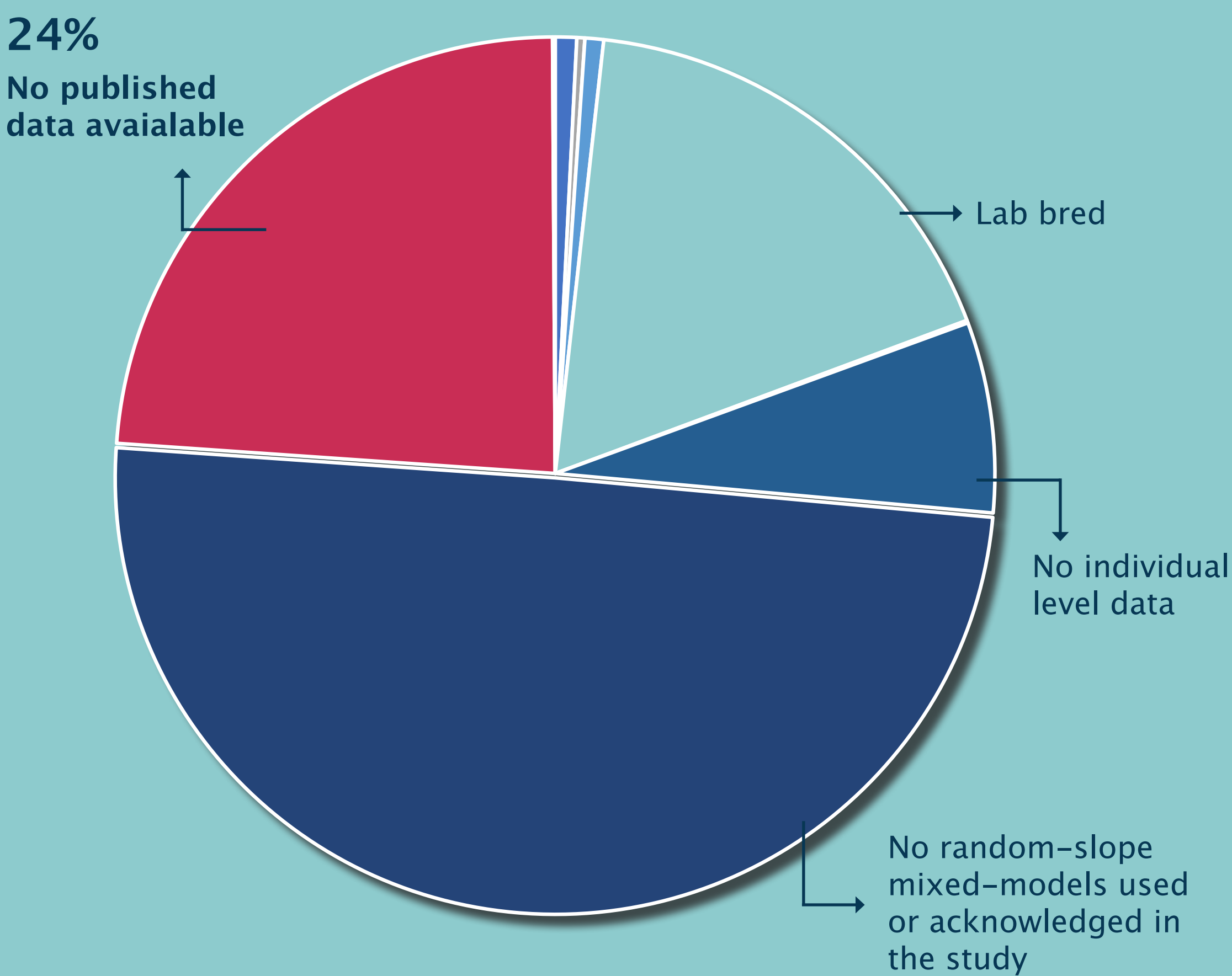
Excluded based on a decision tree
575 records:
• Non-empirical = 349 records
• Human = 101 records
• Domesticated = 47 records
• Lab-bred = 31 records
• Population-level data = 46 records
• Duplicate = 1 record

ELIGIBILITY

FULLTEXT SCREENING

Records screened = 1191 refs
Records after screening = 170 refs

Excluded based on a decision tree
1021 records:
• Non-empirical = 8 records
• Human = 3 records
• Domesticated = 7 records
• Lab-bred = 179 records
• Retracted = 1 records
• No individual-level data = 72 records
• No random slope model = 507 records
• No data available = 243 records
• No access = 1 record



WHAT NEXT?

- Sampling 50 datasets from the selected records
- Re-analysing the data to extract information from random slope mixed models
- Conducting the meta-analysis of the random slope magnitude

